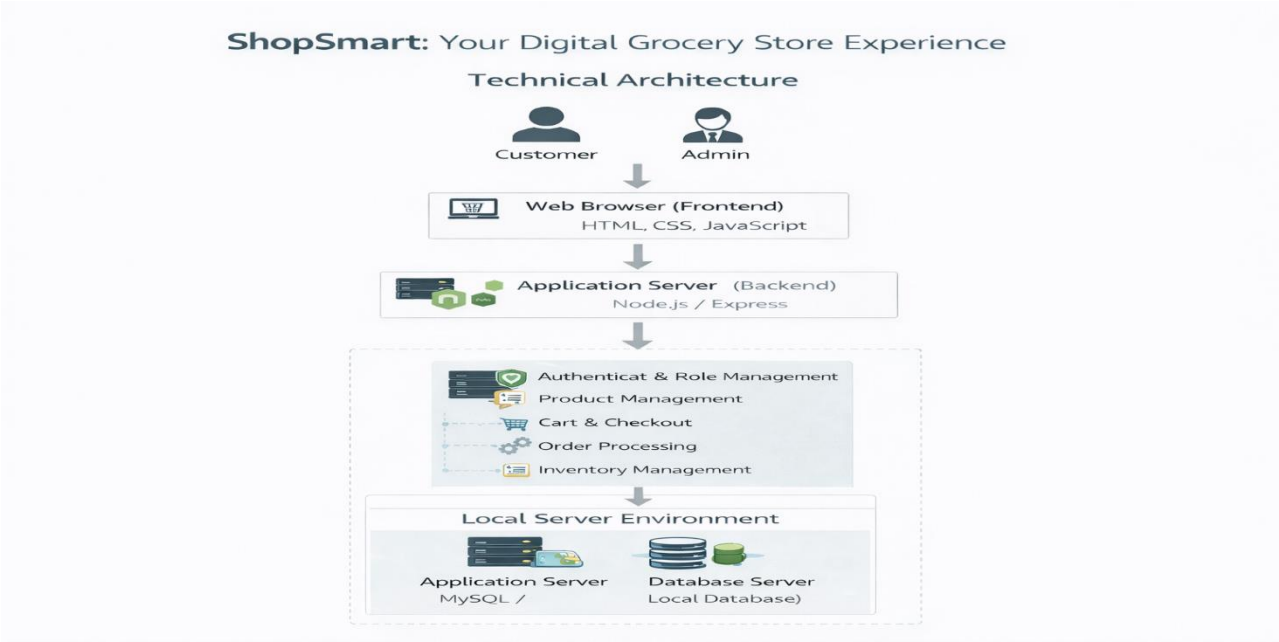
Project Design Phase-II Technology Stack

Date	10 February 2026
Team ID	LTVIP2026TMIDS84782
Project Name	ShopSmart: Your Digital Grocery Store Experience
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: ShopSmart: Your Digital Grocery Store Experience



Guidelines:

1. Application Logic / Technology Blocks

The ShopSmart system includes the following application logic components:

- **User Authentication & Role Management**
  (Customer and Admin login with role-based access)
- **Product Management Module**
  (Add, update, delete products and categories)
- **Cart & Checkout Module**
  (Add to cart, price calculation, order confirmation)
- **Order Processing & Status Management**
  (Order placement, tracking, delivery status updates)
- **Inventory Management Module**
  (Stock updates and availability tracking)

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2. Infrastructure Demarcation (Local)

The system follows a cloud-based architecture.

- **Local Layer:** Customers and Admins access the platform using: Web browser.
- **Server Layer:** Application Server, Order Processing logic, Inventory Management System, Database Server.

3. External Interfaces The system integrates with:

- **REST API Services** – for integration with external enterprise systems (if required)
- **Authentication Services** (JWT/Session-based)– for secure login

4. Data Storage components / services

The system includes:

- **Relational Database**: Stores User details, Product information, Order data, Payment records, Inventory data, Roles & permissions

5. Machine Learning (Future Enhancement)

- Machine Learning is **not implemented** in the current version of ShopSmart.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Web-based platform where customers browse products and place orders	HTML, CSS, JavaScript
2	Application Logic-1	User authentication and role-based access control	Node.js / Express / JWT
3	Application Logic-2	Product management (Add, Update, Delete products & categories)	Backend APIs
4	Application Logic-3	Cart, checkout and order processing logic	Server-side business logic
5	Database	Stores users, products, orders, payments, and inventory data	MySQL / MongoDB
6	Inventory Management	Updates stock availability after order placement	Database operations
7	External API	Payment integration for secure transactions	Payment Gateway API
8	Infrastructure	Application hosted on local server environment	Local Server (Node.js)

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Web Technologies	Frontend and backend implementation	HTML, CSS, JavaScript, Node.js
2	Security Implementations	Role-based access control, authentication.	JWT / Session Authentication
3	3-Tier Architecture	Presentation layer, application layer, database layer	Web + Backend + Database
4	Availability	Runs locally on configured server	Localhost / Server
5	Performance	Efficient product search and order processing	Optimized queries & APIs